Forensic Evidence Portfolio

Coincheck NEM theft, 26 January 2018

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Project: Cryptocurrency Exchange Hack Investigation (independent research project)

1. About this portfolio

This portfolio holds the evidence behind the incident report: the register of every item, the screenshots, the transaction data, the figures and the scripts. The files themselves are in the accompanying folder, 6_Evidence_Files/. Evidence IDs match those used in the report.

I collected the blockchain data on 28 September 2026 from a public NIS1 node and saved it as raw JSON. I then calculated a SHA-256 hash for each file, so anyone can check later that a file has not changed. The article and government screenshots were captured on the same day. All times below are JST (UTC+9) unless marked UTC, using the corrected NEM epoch described in the report.

2. Evidence register

ID	Item	File	Confidence
EVD-001	Reuters Q&A on the Coincheck hack (screenshot). COO and President statements: hot wallet, ~523M XEM, ~3 a.m., noticed 8+ hours later.	screenshots/news_source1.png	Strongly supported
EVD-002	FSA press conference excerpt, 30 Jan 2018, Deputy PM Taro Aso (provisional translation).	screenshots/fsa_notice.png	Confirmed (government record)
EVD-003	CryptoNinjas article carrying the NEM Foundation statement of 28 Jan 2018.	screenshots/ nem_foundation_response.png	Strongly supported
EVD-004	BBC News article on Coincheck's refund pledge.	screenshots/ reimbursement_announcement. png	Strongly supported
EVD-011	Raw transfer records for Address A (Coincheck hot wallet), 4,100 records.	raw/addressA_raw.json	Confirmed
EVD-012	Raw transfer records for Address B (first destination), 250 records.	raw/addressB_raw.json	Confirmed
EVD-013	First-pass combined table of A and B transfers (4,350 rows).	data/transactions.csv	Superseded
EVD-014	First-pass table of the 11 A to B transfers.	data/theft_transactions.csv	Superseded
EVD-015	Screenshot of the live account record for Address A (balance, no co-signatories).	screenshots/ explorer_addressA.png	Confirmed (current state only)
EVD-016	Screenshot of the live account record for Address B (balance 0.001 XEM).	screenshots/ explorer_addressB.png	Confirmed (current state only)
EVD-017	Re-processed table of all 4,350 records: corrected epoch, direction, multisig unwrapped, mosaic flag.	data/transactions_corrected.csv	Confirmed
EVD-018	The 11 transfers from A to B (526,300,010 XEM).	data/ theft_transactions_corrected.csv	Confirmed

ID	Item	File	Confidence
EVD-019	The 8 onward transfers from B (523,000,000 XEM).	data/hop2_sweeps.csv	Confirmed
EVD-020	Collector script used on 28 Sep 2026 (as run, unchanged).	scripts/coincheck_collector.py	n/a
EVD-021	Processing script with the fixes.	scripts/process_raw.py	n/a
EVD-022	Script that finds the A to B and B onward transfers.	scripts/ find_theft_and_sweeps.py	n/a
EVD-023	Figure 1: cumulative XEM moved, A to B and B onward, against reported times.	figures/ fig1_cumulative_movement.png	Analysis
EVD-024	Figure 2: flow of funds from A to B to eight addresses.	figures/fig2_fund_flow.png	Analysis
EVD-025	Reproducibility notes: steps, expected output, method notes and limitations.	REPRODUCIBILITY.md	n/a
EVD-026	Chain of custody: source, method, collector, date, storage and transformations for each item.	CHAIN_OF_CUSTODY.csv	n/a
EVD-027	Script that cross-checks the 19 key transactions against a second NIS1 source (not yet run on a second node).	scripts/ crosscheck_second_node.py	n/a
EVD-028	Finding-to-evidence matrix: each finding with its evidence IDs and processing step.	FINDINGS_MATRIX.csv	n/a

Table 1. Evidence register. Two first-pass files (EVD-013, 014) are kept for the record and are superseded by EVD-017 and 018.

SHA-256 hashes

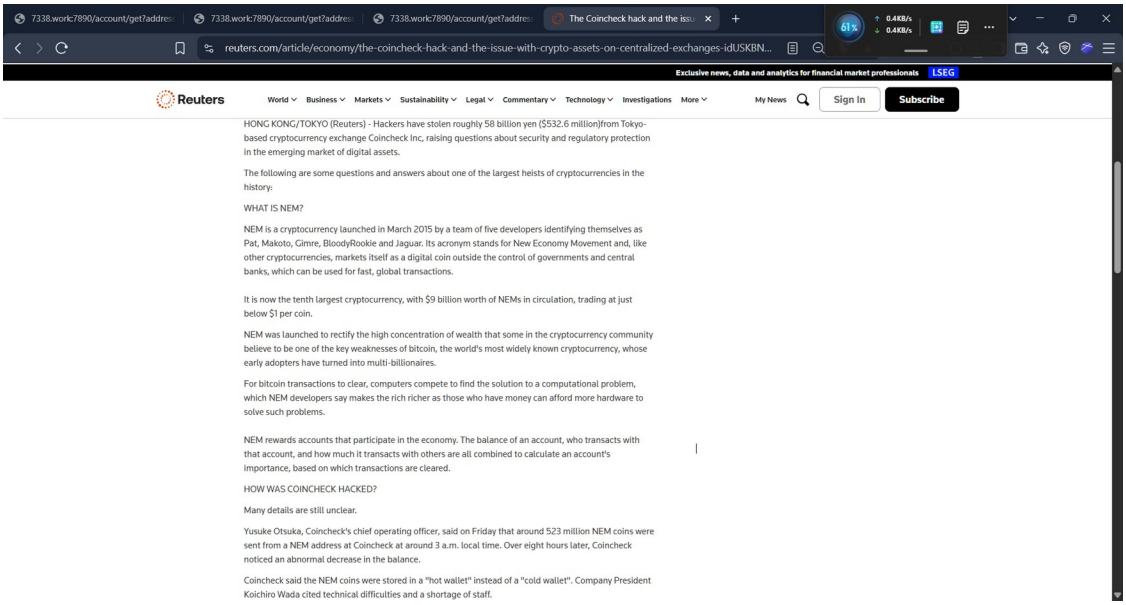
ID	SHA-256
EVD-001	ec0127f24353ecc96db56eeccf19c81ba8c4f0a25c924e049fd4a2ac061388ef
EVD-002	ad4267a630331be3bcc07107c6b52a835b4df64a8e5c645e63f8e539a360a787
EVD-003	8ade9cd8038f21621ff0414ab7b25ddc39ee25eacfd8fe30daa0fb2d8f901aed
EVD-004	e7ea2d0ac0480fff7f40ba92324c0a60d145621acca7c3dd2983559cc64c2f1c
EVD-011	662d1bc39d48f14ec6da63365c3868d1efba88c423d4bef94d7eb403ea853344
EVD-012	9a6c67bbda4f1081448722daf88433f141ea26c958aea13a8c80ab90e1641831
EVD-013	e862fdc910a231da090f9fffca99cc734ef6d56f91778fec896d7863ae016d69
EVD-014	47dab81a3a6d70d067dd2b651c0646ad3b723743cd59bb3974de4ba9c810d32f
EVD-015	336586b3ed110cd67d51599429df3f23f57be8b84b4a5c0f696a7a179b83424b
EVD-016	5aeefaffab0a21c817594a935fac306e2935e7b7372a031b48c188b01aaa1320
EVD-017	d6429cb5033390848fe254b926b7c3689b002a1636e30c6aaf396daf184a2936
EVD-018	bc65946bee862dddf2d656e9817a24741227b79b427295883e5d25b3952965c3
EVD-019	d8a1d52bbf3adb994110a1d78f3fc892565c56ae8992b9cbdc16aa6894f53798
EVD-020	3f11fcd3e51521aab107a8d1ba86d856674a26324d7a6369714402102cb1d098
EVD-021	8999f87d22d68ec644e5b204214a061efe8f63df538b13f965e87746f9dd7bff
EVD-022	21d34ad714857935025a09c5e9af1670141bd2b3b88c3705cc651707949d3c29
EVD-023	f7deb82db49bf2a9dea1db73bdf568509dbfa7979e41979965464d8fe50409a3
EVD-024	8f115c5d5fc80cf03b018537a9c2cef75604e03217ba6bfcc0eb67cd97fc4235
EVD-025	529082d19afae4d022138391d3063a4740db90942dabeba8badede03265972d1

ID	SHA-256
EVD-026	`00436ac8c1bda74c546e8c68b9c5fd56d4612681165a8f3586b5d7c89f6cf919`
EVD-027	`3b0ac347e0e91bb295c10bf7747b114df6432e0f48c9c91123cf9958654e9037`
EVD-028	`10ebff9d9fb3bb0750a125d7a90926c47ea644af82cd159b331cfd34d863aec8`

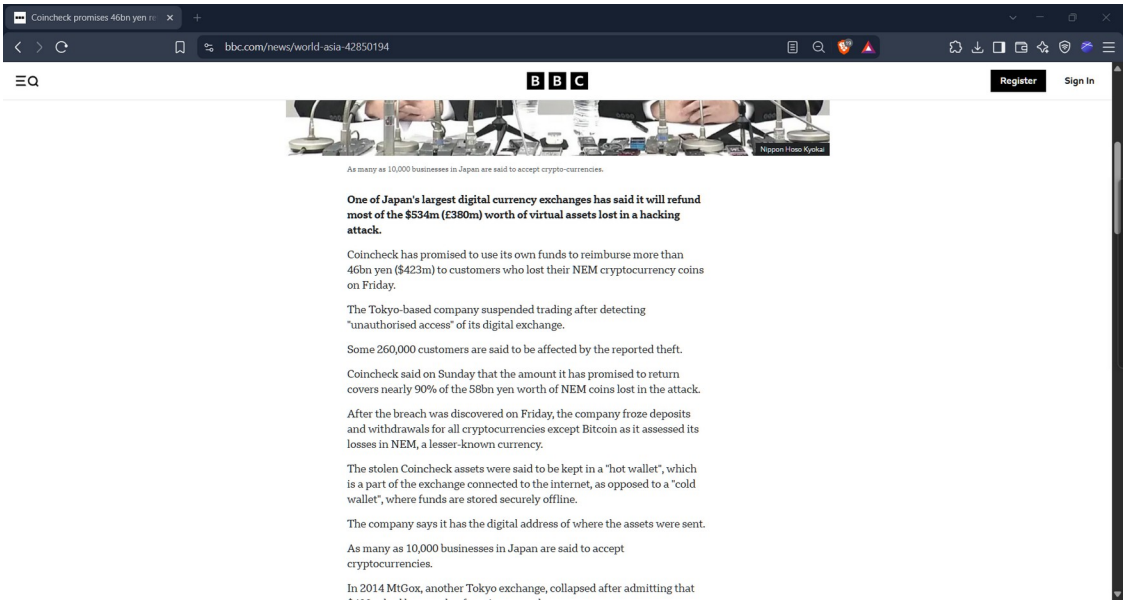
Table 2. Full SHA-256 values. The same list is in EVIDENCE_REGISTER.csv.

3. Screenshots

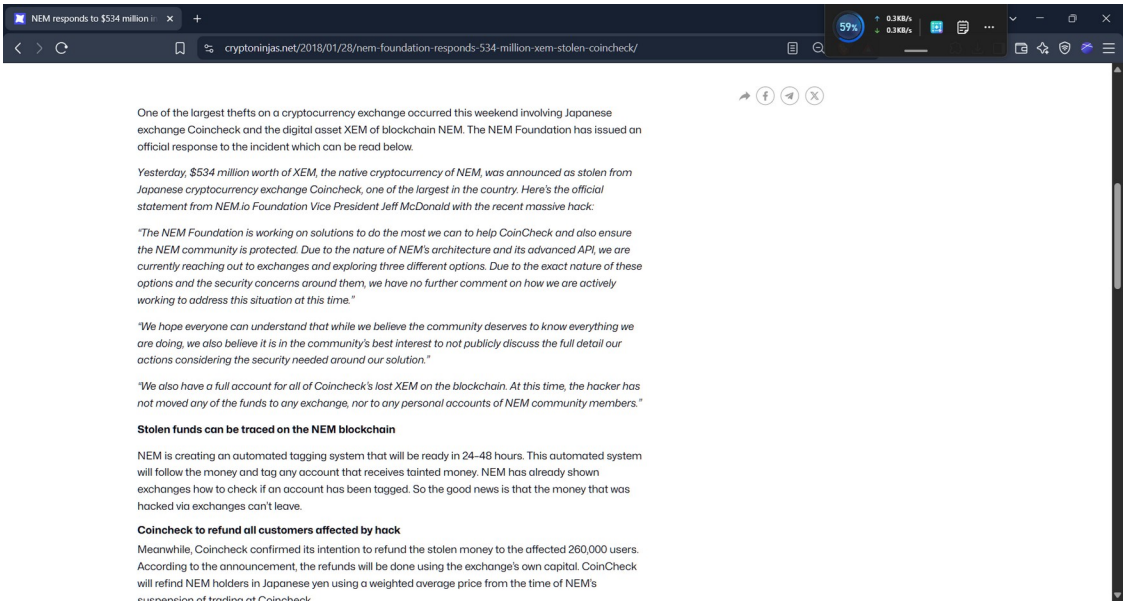
Each screenshot below is listed in the register with its hash. The article screenshots do not all show the publication date, so the URL and source are recorded in the register.



EVD-001. Reuters Q&A. Shows the COO saying about 523 million NEM coins were sent from a Coincheck address at around 3 a.m. and that Coincheck noticed over eight hours later, that the coins were in a hot wallet, and the president citing technical difficulties and a shortage of staff.



EVD-004. BBC News. Shows the $534 million figure, the more than 46bn yen refund pledge, about 260,000 affected customers, the freeze on all cryptocurrencies except Bitcoin, and the hot wallet description.



NEM responds to $534 million in

cryptoninjas.net/2018/01/28/nem-foundation-responds-534-million-xem-stolen-coincheck/

One of the largest thefts on a cryptocurrency exchange occurred this weekend involving Japanese exchange Coincheck and the digital asset XEM of blockchain NEM. The NEM Foundation has issued an official response to the incident which can be read below.

Yesterday, $534 million worth of XEM, the native cryptocurrency of NEM, was announced as stolen from Japanese cryptocurrency exchange Coincheck, one of the largest in the country. Here's the official statement from NEM.io Foundation Vice President Jeff McDonald with the recent massive hack:

"The NEM Foundation is working on solutions to do the most we can to help CoinCheck and also ensure the NEM community is protected. Due to the nature of NEM's architecture and its advanced API, we are currently reaching out to exchanges and exploring three different options. Due to the exact nature of these options and the security concerns around them, we have no further comment on how we are actively working to address this situation at this time."

"We hope everyone can understand that while we believe the community deserves to know everything we are doing, we also believe it is in the community's best interest to not publicly discuss the full detail our actions considering the security needed around our solution."

"We also have a full account for all of Coincheck's lost XEM on the blockchain. At this time, the hacker has not moved any of the funds to any exchange, nor to any personal accounts of NEM community members."

Stolen funds can be traced on the NEM blockchain

NEM is creating an automated tagging system that will be ready in 24–48 hours. This automated system will follow the money and tag any account that receives tainted money. NEM has already shown exchanges how to check if an account has been tagged. So the good news is that the money that was hacked via exchanges can't leave.

Coincheck to refund all customers affected by hack

Meanwhile, Coincheck confirmed its intention to refund the stolen money to the affected 260,000 users. According to the announcement, the refunds will be done using the exchange's own capital. CoinCheck will refind NEM holders in Japanese yen using a weighted average price from the time of NEM's

EVD-003. CryptoNinjas, 28 January 2018. Carries the NEM Foundation vice president's statement about having a full account of the stolen XEM, and the plan for an automated tagging system.



Press Conferences : FSA

fsa.go.jp/en/conference/minister/2018/20180130.html

Provisional Translation

Press Conference by Taro Aso, Deputy Prime Minister, Minister of Finance, and Minister of State for Financial Services

(Excerpt)

(Tuesday, January 30, 2018, 8:38 am to 8:41 am)

[Questions and answers:]

Q. An outflow of funds about 58 billion yen from the quasi-virtual currency broker dealer Coincheck occurred, and the Financial Services Agency (FSA) yesterday issued an order for business improvement to Coincheck. Shortcomings have been pointed out in Coincheck's security system, so could you please tell us how you see the future of virtual currency broker dealers and what kind of oversight approach you think the FSA should adopt?

A. It was very regrettable that a large-scale outflow of virtual currency from Coincheck occurred on last Friday. In the light of the details reported thus far from Coincheck, we issued an order for business improvement to Coincheck requiring them to look into the causes of the incident, adopt measures to prevent recurrence, and take steps to protect their customers. The FSA has been having our IT system experts verify virtual currency broker dealers' IT systems during registration screening and post-registration monitoring. We think that virtual currency broker dealers should further enhance the level of their internal control of their IT systems. Accordingly, we requested that virtual currency broker dealers besides Coincheck make urgent self-check of their systems, and we will consider conducting on-site inspections as needed. In any case, the FSA continues to conduct proper monitoring of virtual currency broker dealers for the sake of protecting users. In addition to this, with regard to the dealing of virtual currency, it is very important to make the right balance between user protection and user convenience.

Financial Policies Monitor

FinTech Support Desk

Financial Market Entry Office

Application and Notification

Subscribing to Information Distribution Service

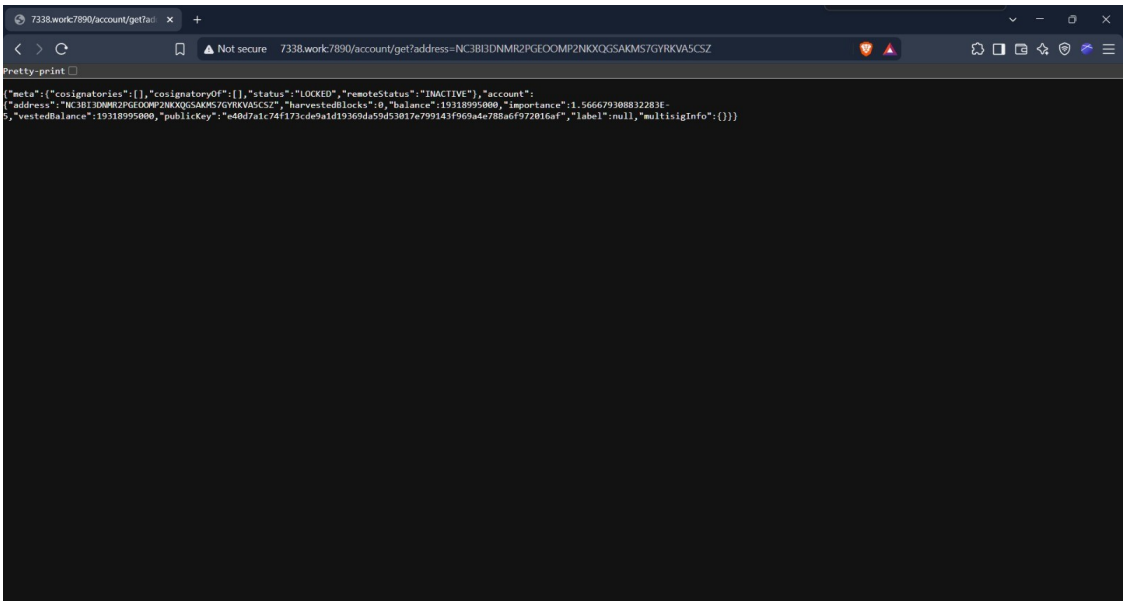
Other Links

Financial Research Center (FSA Institute)

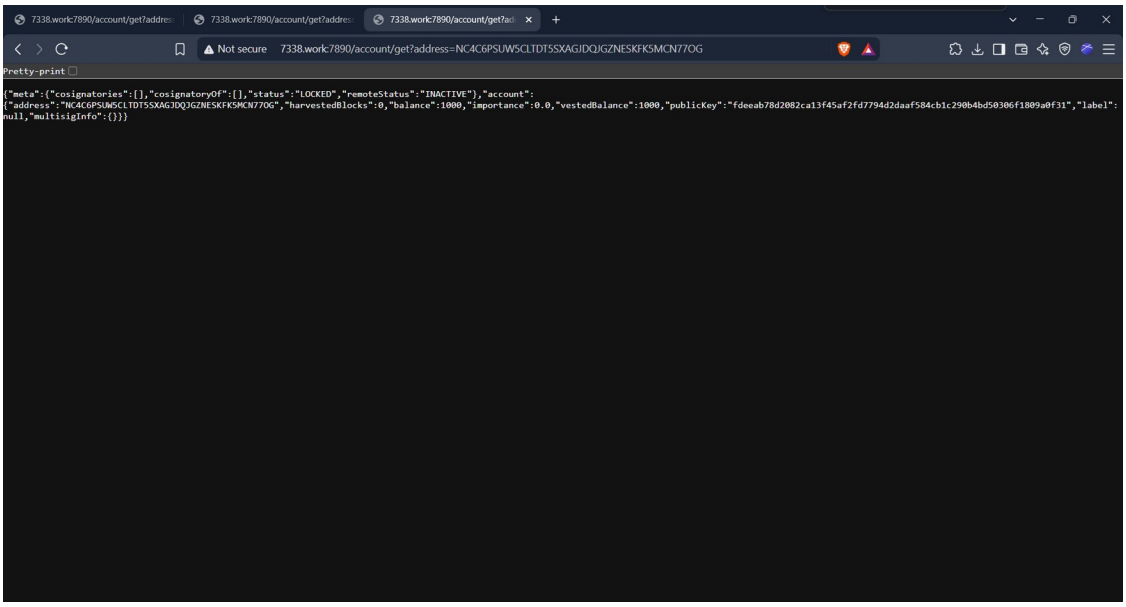
Securities and Exchange Surveillance Commission

Certified Public Accountants and Auditing Oversight Board

EVD-002. FSA press conference, 30 January 2018. Shows the reporter's question and Deputy Prime Minister Aso's answer about the business improvement order. It is marked as a provisional translation.



EVD-015. Address A account record (captured 28 September 2026). Balance 19,318.995 XEM, no co-signatories and no multisig information. This is the current state only.



EVD-016. Address B account record (captured 28 September 2026). Balance 0.001 XEM, no co-signatories. This is the current state only.

4. Transaction data

4.1 Transfers from Address A to Address B

#	UTC	JST	Block	XEM	Transaction hash
1	2018-01-25 15:02:13	2018-01-26 00:02:13	1475043	10	95c508af56b5b3cf40bd01bdeb72e9d316a90b60611e0f1c03860b7e6e5f1494
2	2018-01-25 15:04:56	2018-01-26 00:04:56	1475044	100,000,000	c2921d538a6192bf35e0bb49cefcf6aa8e19acb3fef08f70f5b6b977d9d50aee

#	UTC	JST	Block	XEM	Transaction hash
3	2018-01-25 15:06:46	2018-01-26 00:06:46	1475045	100,000,000	913fb7766fe10b7ca0955f1806772b167d92cd4b1495e9f315f194f3589a8660
4	2018-01-25 15:07:04	2018-01-26 00:07:04	1475045	100,000,000	94cb4ac7c0a45a9678abf8ca087e087c2a813c1b1c2db2dfb12a37387a0c27aa
5	2018-01-25 15:08:21	2018-01-26 00:08:21	1475046	100,000,000	1a4075a308bc965d1b8a3dc62109f70c52f59036f5bde6e80799067536d1404d
6	2018-01-25 15:09:22	2018-01-26 00:09:22	1475047	100,000,000	f05bc90416ba4a435120cff3cef5bbba2c832c48a437ddbfb6dee6d78528cc56
7	2018-01-25 15:10:36	2018-01-26 00:10:36	1475048	20,000,000	282fbe0ca9aca9540adc2a1950144fca8e890c2bffac7e989a9450b2f8292021
8	2018-01-25 15:21:14	2018-01-26 00:21:14	1475058	3,000,000	3c7c835a1d8df946536bb3c0dd23c10c3b02a22220bac3fe3f9b871d26be5424
9	2018-01-25 18:35:09	2018-01-26 03:35:09	1475257	1,500,000	f37bf19fb5533e7bb86696906bf0dde08707cd2a3cb888f518f9cc5019e1163d
10	2018-01-25 19:33:20	2018-01-26 04:33:20	1475309	1,000,000	b74df48a67f93a7224d9b34645c409ee859b9b220551a06c0b68a9fab8e2a38e
11	2018-01-25 23:26:13	2018-01-26 08:26:13	1475553	800,000	2705bbd6b00b0d2d3c1c69d254b0fc2e7af71e1ba353a24fcd736a69a00f585e

Table 3. The 11 transfers, 526,300,010 XEM in total (EVD-018). The first eight add up to 523,000,010 and the last three to 3,300,000.

4.2 Transfers from Address B to eight addresses

#	JST	XEM	Recipient	Transaction hash
1	2018-01-26 02:57:24	30,000,000	NCTWFIOOVITRZYSYIGQ3PEI3IMVB25KMED53EWFQ	77b1a737756aa0bec76a109fbd712639869ed390d68c907cedc0f10258df7600
2	2018-01-26 02:58:42	50,000,000	NA7SZ75KF6ZKK267TRKCJDJBWP5JKIC2HA5PXCKW	c088ad69ea7ef4f3895081e8c3b5495162e2083647a15364574bd3eeb1cba057
3	2018-01-26 03:00:33	50,000,000	NDODXOWEIZGJSMAEURXACF4IEHC2CB7Q6T56V7SQ	1efe1dd9cdb87fafcba2228a36f5dd3b5a29e456d85e49a22688234797be920a
4	2018-01-26 03:02:12	750,000	NBKLQYXEIVEEGARYPUM62UJIFHA3Y6R4LAPU6NP4	73078db7610025490da5bda2c430b35c0c5dca74fd506895137257d5734f8672
5	2018-01-26 03:14:09	100,000,000	NDZZJBH6JZPYSWRPRYHALLWMITWHOYTQGXR53HAW	5d3da55670fb26339484612f2917d10f3f41eea19cafde981f03ceaf7299d43c
6	2018-01-26 03:18:07	100,000,000	NB4QJJCLTZWVFWRFBKEMFOONOZFDH3V5IDK3G524	7d8693ea074a44aafac69ed05cd4bd23b6ad2177605c8f14ac8657e7931fa974
7	2018-01-26 03:28:44	100,000,000	NDDZVF32WB3LWRNG3IVGHCOCAZWENCNRGEZJVCJI	bf46435b77bac25284e1c082acce28ca065ecd650476045650f18dbfc2d8eb23
8	2018-01-26 03:29:54	92,250,000	NA6JSWNF24Y7DVIUVPKRNAY7TPOFJJ7G2URL7KU5	c39cce21c937167b211c7eb104a98e66f1c496103a8f35caea30968bab233dc1

Table 4. The eight onward transfers, 523,000,000 XEM in total (EVD-019).

4.3 Figures

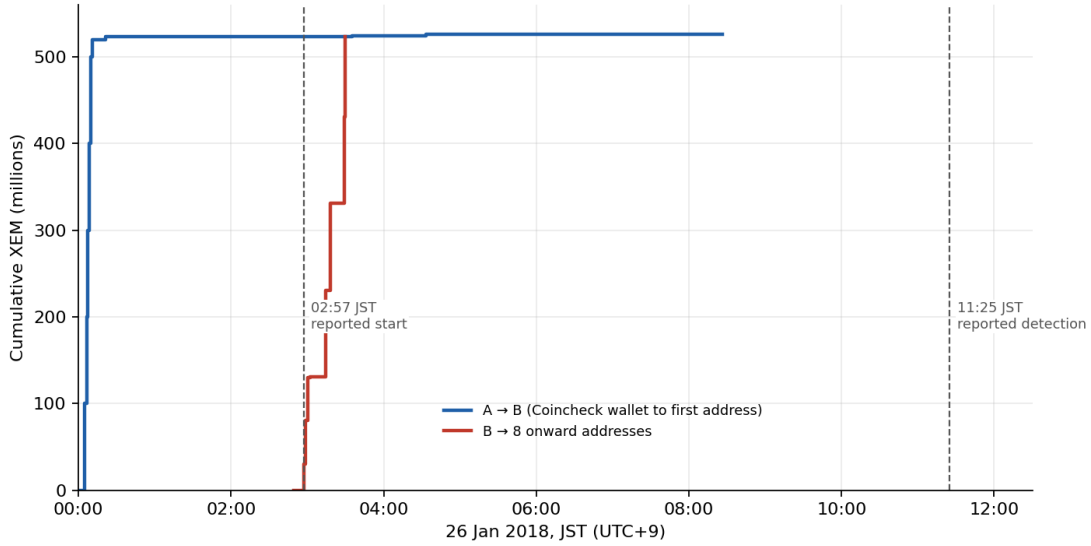


Figure 1. Cumulative XEM moved from A to B and from B onward, against the reported start and detection times (EVD-023).

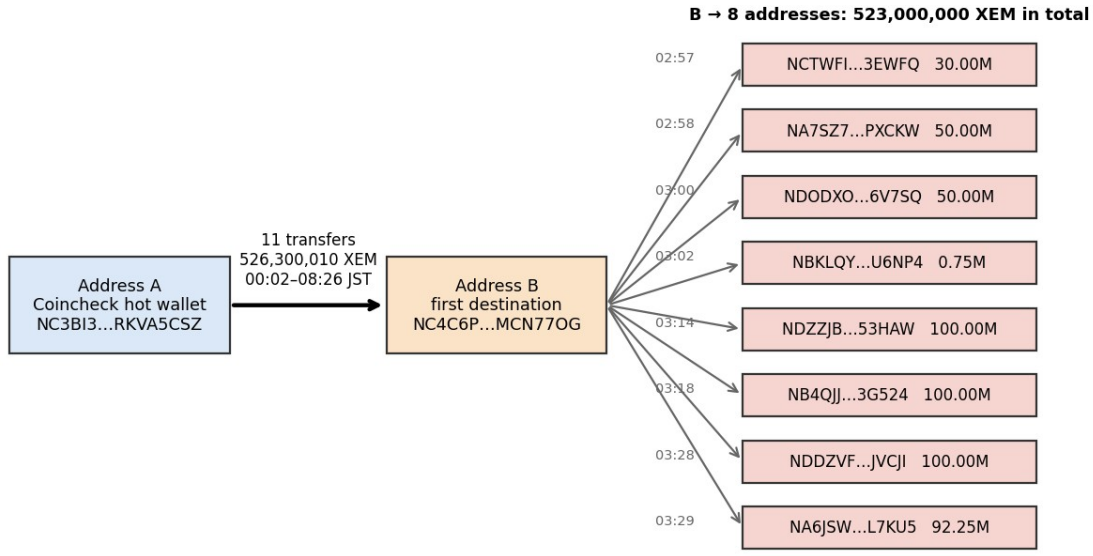


Figure 2. Flow of funds from Address A to B to eight addresses (EVD-024).

5. Reproducing the analysis

1. Run scripts/coincheck_collector.py to download the raw records from a NIS1 node, or use the saved raw/addressA_raw.json and raw/addressB_raw.json.
2. Run python scripts/process_raw.py to build data/transactions_corrected.csv.
3. Run python scripts/find_theft_and_sweeps.py to list the A to B transfers and B's onward transfers. The expected output is 11 transfers (526,300,010 XEM) and 8 onward transfers (523,000,000 XEM).

4. Optional: run python scripts/crosscheck_second_node.py --node <second NIS1 node URL> to compare the 19 key transactions with another node. This has not yet been run.

6. Data quality notes

- The first collector script used an epoch of 00:00:00 UTC, so its times are 6 minutes 25 seconds early. process_raw.py uses 00:06:25 UTC.
- 138 of 4,350 records are multisignature-wrapped and 39 carry mosaics. None of the transfers used in the report is one of them.
- The current balances in the two account screenshots describe the present, not 2018.
- Data came from a single node and should be cross-checked against a second node. scripts/crosscheck_second_node.py (EVD-027) is ready for this check; it has not yet been run.
- The eight onward transfers (EVD-019) were selected as plain transfers of 500,000 XEM or more from B before 26 January 00:00 UTC. B also made 37 smaller later transfers totalling about 3.3 million XEM, which are not traced.

7. Chain of custody

CHAIN_OF_CUSTODY.csv, in the evidence folder, records for every item where it came from, how it was acquired, who collected it, when, where it is stored and what was done to it afterwards. Table 5 summarises it. Kshitij Randhire, the author, collected every acquired item. Derived files were produced by the scripts named in the table.

Items	Source or endpoint	How acquired	Collected	Transformation
EVD-001 to 004	News and government pages (URLs in the register)	Browser screenshot taken manually	28 Sep 2026	None recorded
EVD-011, 012	NIS1 node 7338.work:7890, /account/transfers/all, 100 records per page	scripts/coincheck_collector.py (EVD-020)	28 Sep 2026	Saved by the script as formatted JSON; values unchanged
EVD-015, 016	7338.work:7890/account/get?address=… for A and B (URL visible in the screenshot)	Browser screenshot taken manually	28 Sep 2026	None recorded; current state only
EVD-013, 014	Derived from EVD-011, 012	First-pass processing	Not recorded	Epoch error (6 min 25 s early); superseded
EVD-017 to 019	Derived from EVD-011, 012	scripts/process_raw.py, then scripts/find_theft_and_sweeps.py	Not recorded	Corrected epoch, direction, multisig unwrap, mosaic flag; then filters
EVD-023, 024	Derived from EVD-018, 019	Figures	Not recorded	None
EVD-020 to 022, 025 to 028	Written for this project	Authored	Not recorded (EVD-020 run on 28 Sep 2026)	EVD-025 edited before publication; its hash is for the published version

Table 5. Chain of custody summary.

Not recorded: the time of day of each collection and the exact command line used. They are marked as not recorded in CHAIN_OF_CUSTODY.csv instead of being reconstructed from memory. For any later collection, record the UTC time, the full command and the node address at the moment of collection.

8. Finding-to-evidence matrix

Each major conclusion is mapped to its evidence IDs and the processing step behind it, so a reader can verify it without searching the five deliverables. FINDINGS_MATRIX.csv holds the same table with a column saying how to verify each finding.

ID	Finding	Report	Evidence	Processing step	Confidence
F-01	526,300,010 XEM moved from A to B in 11 transfers, the first at 00:02:13 JST	7.2	EVD-011, 017, 018	collector, process_raw.py, find_theft_and_sweeps.py	Confirmed
F-02	B sent exactly 523,000,000 XEM to eight addresses, 02:57:24 to 03:29:54 JST	7.3	EVD-012, 017, 019	find_theft_and_sweeps.py (first sweep filter)	Confirmed
F-03	All 11 transfers were single-signer plain transfers (type 257)	5, 7.2	EVD-011, 017, 018	process_raw.py (type, multisig unwrap)	Confirmed for the transfers
F-04	XEM in a hot wallet; president cited technical difficulties and staff shortage	5	EVD-001, 004	None (source statements)	Strongly supported
F-05	About 260,000 customers affected; more than 46bn yen refund pledged	6	EVD-004	None (source statement)	Strongly supported
F-06	FSA issued a business improvement order (29 Jan date inferred)	8	EVD-002	None (source statement)	Confirmed (order); date inferred
F-07	DISC-001: Coincheck's ~523M matches the first eight transfers; last three (3,300,000 XEM) not reflected	7.4	EVD-001, 018	find_theft_and_sweeps.py (first 8 and last 3 sums)	Arithmetic Confirmed; explanation Interpretation
F-08	DISC-002: reported ~02:57 start matches the first onward transfer from B, not the first from A	7.5	EVD-001, 018, 019	process_raw.py (corrected epoch), then time comparison	Interpretation
F-09	Detection gap about 8 h 28 min from the stated start, about 11 h 23 min from the first transfer	3, 5	EVD-001, 018 and a secondary source for 11:25	Time arithmetic on corrected JST times	Confirmed (chain times); strongly supported (11:25)
F-10	First collector used the wrong epoch, times 6 min 25 s early; corrected	1, 7.7	EVD-013, 014, 017, 018, 020, 021	process_raw.py	Confirmed
F-11	Not established: access method, attacker, customer data exposure, onward flow of funds	4, 6.2, 9	None	None	Unknown

Table 6. Finding-to-evidence matrix.